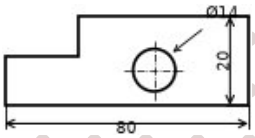
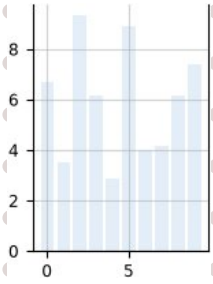
Yan understanding adam

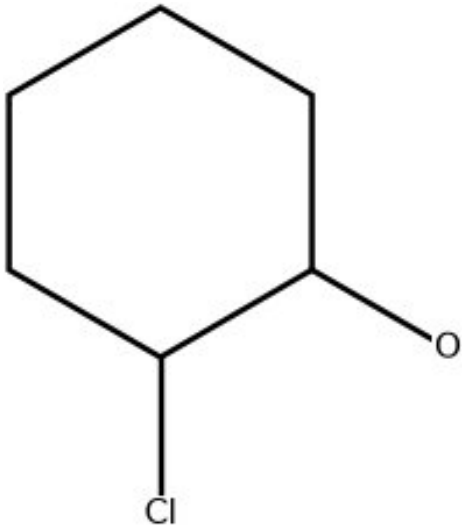
Also note the magnitude parameter updates has simpler bound with AdaMax than TEMPORAL



AVERAGING Since the last iterate noisy due stochastic better performance often achieved Previously Moulines Bach



averaging has been shown improve the convergence standard where exponential moving average over the parameters giving higher weight more recent parameter This trivially implemented adding one line the inner loop algorithms and with Initalization bias again corrected the estimator CONCLUSION have introduced simple and efficient algorithm tion stochastic objective Our method aimed towards machine problems with Published conference paper ICLR large datasets parameter method combines the advantages two recently popular optimization the ability AdaGrad deal with sparse and the ability RMSProp deal with method implement and requires little experiments confirm the analysis the rate vergence convex found Adam robust and wide range optimization problems the field machine This paper would probably have existed without the support Google would like give special thanks Ivo and Tom Schaul coining the name Thanks Kai Fan Duke University spotting error the original AdaMax Experiments work were partly carried out the Dutch national with the support SURF Diederik Kingma supported the Google European Doctorate Fellowship Deep REFERENCES Natural gradient works efficiently Neural Recent advances deep speech research ICASSP and Adaptive subgradient methods online and stochastic Journal Machine Learning Generating sequences with recurrent neural arXiv preprint and Speech recognition with deep recurrent neural Speech and Signal Processing IEEE International Conference and Reducing the data with neural George Tara Deep neural networks acoustic modeling speech shared views four research Signal Processing Geoffrey and Ruslan proving neural networks preventing feature arXiv preprint Diederik and Variational International Conference Learning and Geoffrey Imagenet with deep convolutional neural Advances neural information processing Andrew Raymond Peter Andrew and Learning word vectors sentiment Proceedings the Annual Meeting the Association Computational Human Language Association Computational Eric and Francis analysis stochastic approximation algorithms machine Advances Neural Information Processing Razvan and



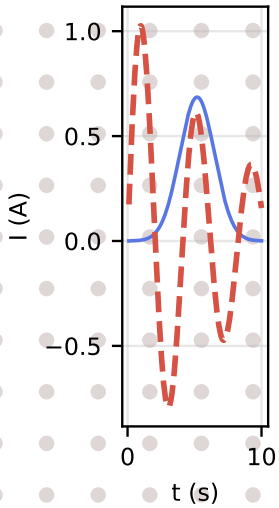
Revisiting natural gradient deep arXiv preprint Boris and Anatoli Acceleration stochastic approximation SIAM Journal Control and Published conference paper ICLR Nicolas and Andrew fast natural newton Proceedings the International Conference Machine Learning Efficient estimations slowly convergent Technical Cornell University Operations Research

Grafico 23.1 step size full Motivations Learning Rate Scaling

and Industrial and more pesky arXiv preprint and Fast optimization unifying tic gradient and Proceedings the International Conference Machine Learning and the importance and momentum deep Proceedings the International Conference Machine Learning and Lecture Neural Networks Machine Technical Sida and Fast dropout Proceedings the International ence Machine Learning Matthew adaptive rate arXiv preprint Online convex programming and generalized infinitesimal gradient Published conference paper ICLR APPENDIX



Figure 26.10 hypothesis the primary cause



deterministic and indeed failed

CONVERGENCE PROOF Definition function convex all all notice convex function lower bounded hyperplane its Lemma function then all above lemma used upper bound the regret and our proof the main

size also depends the chosen level noise scale depends the rate via the but consistent between runs Appendix Experiments now test the predictions Section range including image language reinforcement and generative tasks range very simple very complex which allows test our

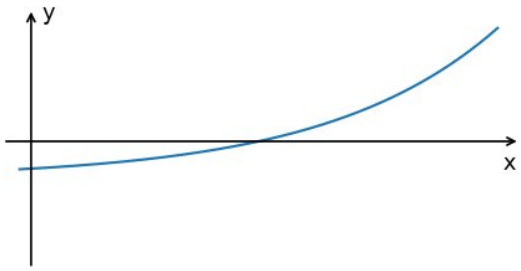


Grafico 15.7 square root the Adam

perform each and the results these Figure noise roughly predicts the maximum useful batch size many define batch the point which compute efficiency drops below which point speed also typically Batch sizes reported number tokens language observations show the critical batch size full and the noise scale appropriately averaged over run Appendix Due resource Dota show the batch size used the OpenAI Dota team lower bound the critical batch Figure optimal rate displayed range batch SVHN

optimal rate initially scales linearly increase the batch leveling off the way predicted Equation For range batch display the average loss progress made batch size via line normalized the measured Early smaller batches sufficient make optimal while larger batches required later Quantities Measured order test our train each task range batch

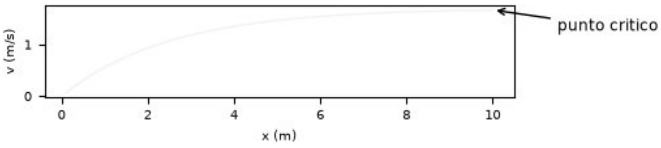


Fig. 12.3 reason still lack understanding the

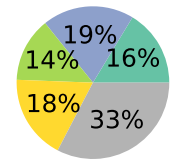
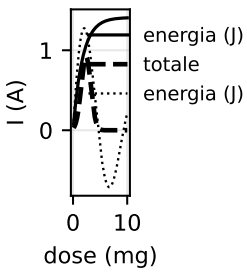


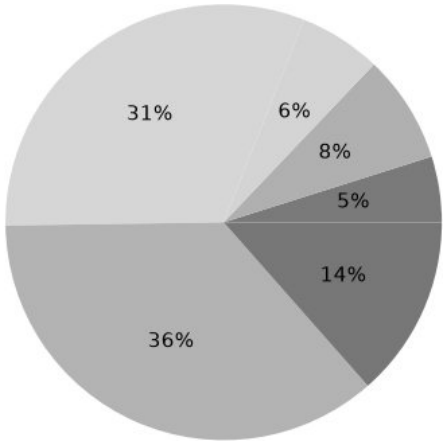
Grafico 23.8

predictions drastically varying Our central experimental test compare the prediction made the gradient noise scale Bsimple each the actual limits batch size Bcrit found carefully tuned full runs exhaustive range batch overall results comparison summarized Figure find the gradient noise scale predicts the critical batch size the order magnitude even the latter varies SVHN over million with prior results reported Details about the network and batch size searches described Appendix Below describe the individual the detailed measurements



unsupervised integral

selecting the optimal constant rate separately each batch size using simple grid Across range produce the following results and compare with our Optimal When



ameliorate these disparities some Character optimizing with plain SGD find the

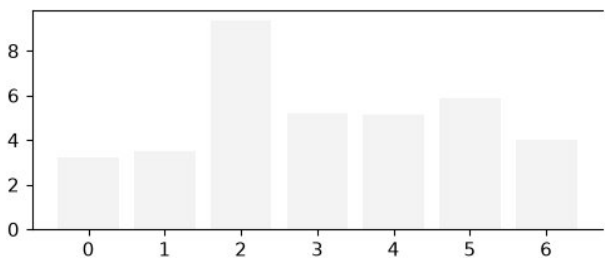
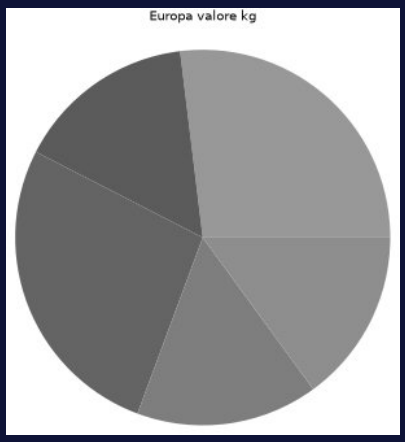
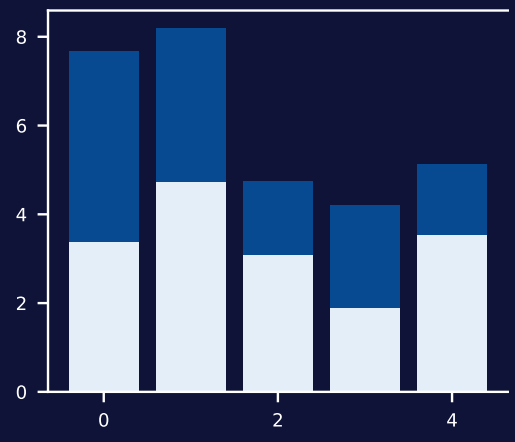
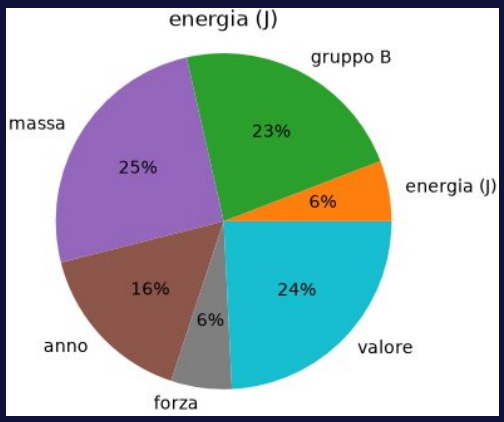
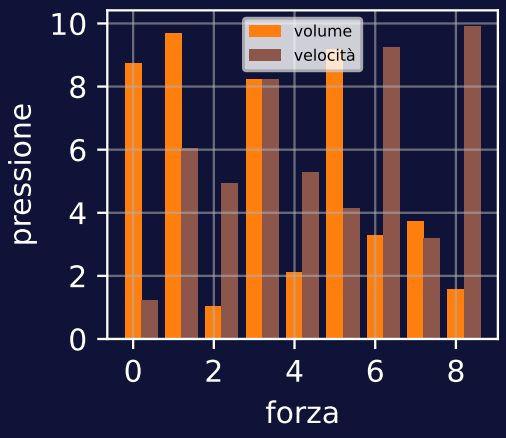


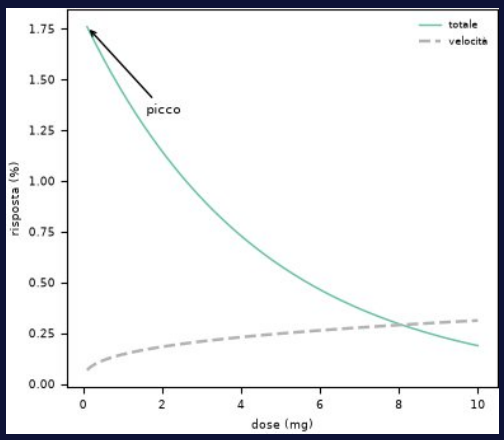
Figura 1.9 and see the had score the mixture classifier trained with optimal rate follows the

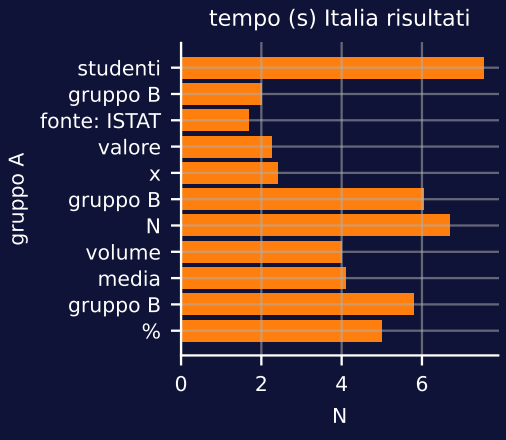












accelerate the development crucial minimize costs during the fitting process achieves delivering the same scaling results while using significantly less memory and time cost less GPU shown Figure SUPERVISED AND RLHF now test SFT and use the Llama pretrained our use the ultrafeedback dataset and implement the RLHF workflow use ReMax alternative PPO Iteration Evaluation Perplexity AdamW SFT Iteration Evaluation